

effort [8, 12]. Optimization-based jailbreaks optimize the tokens input to the LLM, recognized for their effectiveness but requiring extensive computational resources and being often uninterpretable to humans [53, 29].

3 From LLM Safety Landscape to VISAGE Safety Metric

The model landscape is a crucial tool for interpreting model behaviors and understanding model characteristics. Perturbing a model along random directions reveals the local behavior of the model, while interpolating the parameters between two models illustrates the transition process from one model to the other. In this section, we introduce the notion of LLM safety landscape in both 1D (Sec. 3.1) and 2D (Sec. 3.2) scenarios. Sec. 3.3 presents the safety landscape of four popular open-source LLMs and Sec. 3.4 introduces “LLM safety basin” concept and proposes the VISAGE safety metric based on our landscape analysis.

3.1 1D Safety Landscape

Denote θ as the initial LLM model weights. The safety landscape is plotted by perturbing θ along a certain direction $\widehat{d}_1$ and evaluate the new model weights with a single model safety metric:

$$f(\alpha) = \mathcal{S}(\theta + \alpha \widehat{d}_1) \quad (1)$$

where $\mathcal{S}$ is the safety metric defined for a single model, and α is a scalar parameter. For 1D-interpolation, we pick two sets of model weights θ and θ' and the direction is defined by the line connecting these two points, $\widehat{d}_1 = \theta' - \theta$. For 1D-random, θ is the center point and we randomly sample a direction d_1 from Gaussian distribution. We apply layer normalization to d_1 to exclude the effect of scale invariance [31] so that the flatness and the sharpness across different landscape plots are comparable. Specifically, d_1 is normalized to a unit direction and then multiplied by the Frobenius norm of each layer i :

$$\widehat{d}_{1i} = \frac{d_{1i}}{\|d_{1i}\|} \|\theta_i\| \quad (2)$$

In the rest of the paper, we will use 1D-random θ and 1D-interpolation $\theta \rightarrow \theta'$ to represent the above two types of 1D directions.

3.2 2D Safety Landscape

Similar to the 1D landscape, the 2D landscape requires two directions $\widehat{d}_1$ and $\widehat{d}_2$ and the safety landscape is defined as:

$$f(\alpha, \beta) = \mathcal{S}(\theta + \alpha \widehat{d}_1 + \beta \widehat{d}_2) \quad (3)$$

For 2D random, since both directions are randomly sampled from Gaussian distribution, the cosine similarity between $\widehat{d}_1$ and $\widehat{d}_2$ is $\sqrt{2/(\pi n)}$ [15], where n is the dimension of θ . Given current LLMs have billions of parameters, the two random directions are orthogonal and we only perform layer normalization as in Eq. 2. For 2D interpolation, we pick three sets of model weights θ , θ' , and θ'' and compute the interpolated directions $d_1 = \theta' - \theta$, $d_2 = \theta'' - \theta$. Since there is no guarantee that two interpolated directions are orthogonal, we use Gram-Schmidt algorithm to find the orthogonal basis:

$$\widehat{d}_1 = d_1, \widehat{d}_2 = d_2 - \frac{d_1^T d_2}{\|d_1\|^2} d_1 \quad (4)$$

To ensure the scale equivalence of two directions, we rescale $\widehat{d}_2$ to $\left(\frac{\|\widehat{d}_1\|}{\|\widehat{d}_2\|}\right) \widehat{d}_2$. 2D-interpolation landscape is useful when analyzing two finetuned model weights, which are all initialized by the same aligned LLM. In the rest of the paper, we will use 2D-random θ and 2D-interpolation $\theta \rightarrow \theta' \& \theta''$ to represent the above two types of 2D directions.

3.3 Safety Landscape of Open-source LLMs

We show the safety landscapes of four popular open-source LLMs: LLaMA2-7B-chat [44], LLaMA3-8B-instruct [2], Mistral-7B-instruct-v0.2 [27], and Vicuna-7B-v1.5 [10]. For each perturbed model

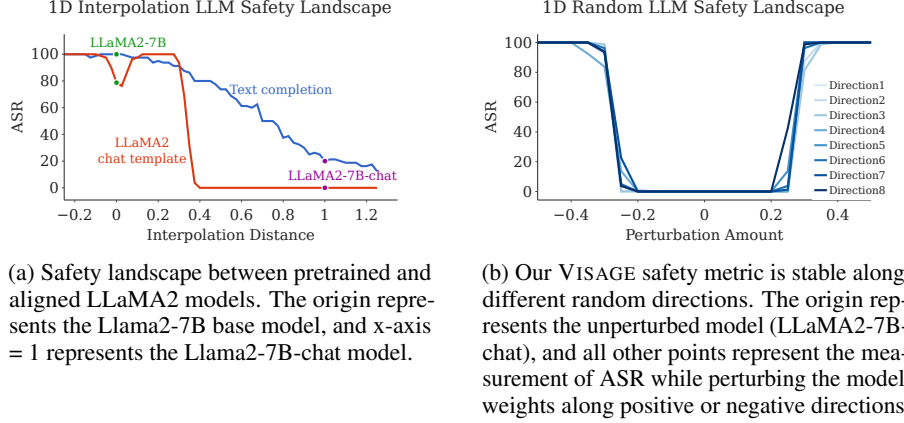


Figure 2: LLM safety landscape: (a) 1D-interpolation LLaMA2-7B $\rightarrow$ LLaMA2-7B-chat safety landscape. When given two models varied by fine-tuning, we utilize linear interpolation to visualize the changes between them. While interpolating the model weights between the base and the chat model, we need to ensure the chat format remains consistent. Thus, we ablate on both chat formats: text completion (no template) and LLaMA2 chat template. The chat model exhibits higher safety than the base model as expected. The base model also shows an increase in safety while using the LLaMA2 chat template. (b) 1D-random LLaMA2-7B safety landscape sampled over different random directions. When provided with a single model, we sample a random normalized direction to visualize its local variations along both positive and negative directions.

along the landscape direction, we evaluate on the first 80 prompts of AdvBench [53] “Harmful Behaviors” split (Adv 80) with ASR as the safety metric. The ASR is measured by refusal keyword detection following the original AdvBench evaluation protocol. Note that S can be any harmfulness evaluation metric, *e.g.*, LLM Judge [52] or Llama Guard [26]. Since a recent user study [37] shows that GPT-4 Judge and refusal keyword detection perform closely on flagging harmful content, we use keyword detection as it is the fastest. We interpolate 20 steps on each axis for all landscapes. To ensure deterministic results, we set top-p as 0 and temperature as 1 [17].

Safety landscape between pretrained and aligned LLMs. Pretraining is the initial phase of training an LLM, aimed at developing a broad understanding of language and knowledge [11, 5]. Alignment, on the other hand, focuses on training LLMs to better follow instructions in prompts and align their behaviors with human preferences [35]. Since the pretrained model is not designed with safety in its first priority, it lacks safety guardrails. In contrast, the aligned model is expected to refuse to respond to harmful user queries. We use LLaMA2 as an example and show the safety landscape of 1D-interpolation LLaMA2-7B $\rightarrow$ LLaMA2-7B-chat in Fig. 2a. Since the pretrained (LLaMA2-7B) and the aligned (LLaMA2-7B-chat) models use different chat templates, both templates are evaluated to ensure the change of safety is due to alignment. Notice that the chat template difference does not exist when comparing the aligned and the finetuned models in later sections as all of them share the same chat template and system prompt. Both lines in Fig. 2a show that the ASR of the aligned model is significantly lower than the pretrained model as expected. Notice that the pretrained model has a less than 100 ASR when using the aligned model chat template. This is because the pretrained model repeats the system prompt in the aligned model’s chat template, which is captured by the keyword detector and treated as successful refusal. When using the aligned model’s chat template, we find that the local region of the aligned model shows the same level of safety. This is surprising because early work has shown that finetuning can easily break LLM’s safety alignment, which may imply the aligned model is a cusp, *i.e.*, sharp corner, on the safety landscape.

Safety landscape of an aligned LLM. Inspired by our findings in the interpolation direction above, we are curious whether this flat safety region exists in other directions for different LLMs. Fig. 1 (top) plots the 1D and 2D random landscape of four popular open-source LLMs. For each LLM, we use the default system prompt provided by the model, with details in Appendix A. We discover that each aligned LLM serves as a robust anchor point, maintaining safety within its local region, but the safety is completely compromised outside of this local region, and the change of safety as steep as a step function. We term this new phenomenon observed universally in the LLM parameter space as “safety basin”. All four investigated LLMs exhibit such phenomenon, but the differences

are the depth and width of the basin. The finding of safety basin generalizes to different evaluation metrics and other safety datasets (Appendix B), and the model still generate fluent output when ASR is high (Appendix D). Besides, we also find that a larger model side exhibits a wider safety basin (Appendix E).

3.4 VISAGE Safety Metric

The landscape visualization and analysis suggest that the average depth of the safety basin can serve as a good indicator for measuring LLM safety, reflecting both the safety of the original aligned model and the robustness of the model when its parameters are perturbed. Formally, for an n -D random safety landscape, we define VISAGE safety metric as the average safety margin of all models we have sampled along all random directions:

$$\text{VISAGE} = \mathbb{E}_{\alpha \sim \mathcal{U}(-a, a), \beta \sim \mathcal{U}(-b, b), \dots} [S_{max} - \mathcal{S}(\alpha, \beta, \dots)], \text{ s.t. } \mathcal{S} < S_{max} \quad (5)$$

where α and β are all sampled from uniform distribution and we use $a = b = 0.5$ as we find that LLMs are completely broken after perturbing more than half of its norm. $\mathcal{S}$ is a monotonically decreasing function in terms of safety, as a lower $\mathcal{S}$ means a safer model. S_{max} is the maximum possible value for $\mathcal{S}$. When ASR is used as the safety metric, $S_{max} = 100$.

Stability of VISAGE. Since the definition of VISAGE involves random directions, we run a stability test to find out how many sampled directions the model converge to a stable value. We take 1D-random LLaMA2-7B-chat as an example and show the result in Fig. 2b. We sample 8 different directions until the average converges and find that the average of 3 different directions is close enough to the final average. Thus, we use the mean of VISAGE along three different directions as the evaluation metric in the rest of the paper. In Fig. 1, we also compute VISAGE for both 1D and 2D random landscape and the ranking of those two dimensions remain the same. Thus, we use 1D VISAGE for faster evaluation.

We compute the VISAGE score for all four LLMs using their default system prompts and chat templates. Since LLaMA3-8B-instruct does not have a default system prompt, we use the LLaMA2 system prompt. The VISAGE ranking is as follows: LLaMA3-8B-instruct > LLaMA2-7B-chat > Mistral-7B-instruct-v0.2 > Vicuna-7B-v1.5. We also evaluate these four models on all 520 prompts of AdvBench ‘‘Harmful Behaviors’’ split (Adv 520). The ASR of the models are as follows: LLaMA3-8B-instruct (0.38), LLaMA2-7B-chat (0.19), Mistral-7B-instruct-v0.2 (1.15), and Vicuna-7B-v1.5 (2.5). Although the ASRs of all four LLMs are close, the VISAGE reflects the safety of a model’s local region, indicating the risk after finetuning, which we will explore in the next section.

4 Why can simple finetuning easily break LLM’s safety alignment?

In this section, we navigate the LLM safety landscape and explore why safety alignment can be easily compromised by finetuning with only a few adversarially designed training examples. Sec. 4.1 details the finetuning settings. In Sec. 4.2, we discover that different LLMs have varying rates of vulnerability to finetuning, and our task agnostic VISAGE safety metric measures the risks in finetuning without assumptions on the finetuning dataset. In Sec. 4.3, visualizing the safety landscape of the aligned model enables us to understand, for the first time, how finetuning compromises safety by dragging the model away from the safety basin. Though finetuning can easily break the safety alignment, we demonstrate that as long as the finetuning process stays within the safety basin, the safety of the finetuned model remains intact in Sec. 4.4.

4.1 Finetuning settings

We finetune on the harmful samples created by Qi et al. [37], which were sampled from Anthropic red-teaming dataset [14]. Following the standard OpenAI finetuning API [36], each training sample is structure in a one-round conversation format.

We ensure that the system prompt used during finetuning remains consistent with the aligned model so that the differences in safety are indeed induced by finetuning. We adhere to the official finetuning recipe¹ and conduct full parameter finetuning. Following the training hyperparameters in Qi et al.

¹<https://github.com/facebookresearch/llama-recipes>